ISMF

IRISH STUDENT MANAGED FUND

Position Sizing and Risk Management in Systematic Trading



ISMF Algo-Trading Bootcamp
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Learning Outcomes

- Translate a trading signal into a target position.
- Scale exposure using estimated volatility.
- Apply leverage, concentration and liquidity limits.
- Monitor drawdowns and evaluate portfolio stress.



From Signal to Position

Our model generates a buy signal.

- Should we buy €1,000, €10,000 or €100,000?
- How volatile is the asset?
- What risks are already present in our portfolio?

A sizing rule turns a signal into a target position.



Our Running Example

Input	Value
Portfolio Equity	€100,000
Instrument	Stock A
Current share price	€50
Trading signal	Buy
Current holding	0 shares
Decision frequency	Daily

How do we turn this buy signal into a number of shares?



From Signal to Target Weight

We start with a simple rule: allocate 10% of portfolio equity when the signal indicates Buy.

Signal	Target weight
Buy	+10%
Neutral	0%
Short	-10%

Target Exposure = Portfolio Equity × Target Weight
€100,000 × 10% = €10,000.

Our initial target exposure is €10,000.



From Target Exposure to Shares

From our running example:

- Target exposure: €10,000
- Share price: €50

Target Shares = Target Exposure ÷ Share Price

€10,000 ÷ €50 = 200 shares

So our Target position is 200 shares of Stock A.



Target Position versus Required Trade

Required Trade = Target Shares – Current Shares.

Current Holding	Target Holding	Required Trade
0 shares	200 shares	Buy 200 shares
150 shares	200 shares	Buy 50 shares
250 shares	200 shares	Sell 50 shares
200 shares	200 shares	No trade

A buy signal can still require a sell order if we already hold more than the target.



Equal Capital ≠ Equal Risk

Consider two stocks with the same allocation but different volatility.

	Stock A	Stock B
Position value	€10,000	€10,000
Estimated annualised volatility	20%	40%
Estimated annualised P&L standard deviation	€2,000	€4,000

P&L Standard Deviation ≈ Position Value × Return Volatility

With equal capital invested Stock B has twice the estimated standalone volatility risk.



Sizing Positions using Volatility

Adjust position size to target a chosen level of standalone volatility risk.

Target weight = Signal × Risk target ÷ Volatility

- Signal: +1 for Buy, 0 for Neutral, -1 for Short.
- Risk Target: Desired standalone P&L volatility as a percentage of portfolio equity.
- Volatility: Estimated volatility of the asset's returns

Higher estimated volatility → smaller position.

Lower estimated volatility → larger position.



Volatility Based Sizing: Example

Portfolio Equity = €100,000 | Annual risk target per position = 2%

Both signals = +1 (Long)

	Stock A	Stock B
Estimated annualised volatility	20%	40%
Target Weight	2% ÷ 20% = 10%	2% ÷ 40% = 5%
Target Exposure	€10,000	€5,000
Annual P&L standard deviation*	**€2,000**	**€2,000**

*Estimated, assuming fixed Euro exposure. Not a maximum loss.

Halving the exposure offsets twice the estimated volatility.



Estimating Volatility

1: Calculate daily returns:

Daily return = Today's closing price ÷ Yesterday's closing price – 1.

2: Calculate their rolling standard deviation:

Use the most recent 60 daily returns as an illustrative lookback window.

3: Annualise the estimate:

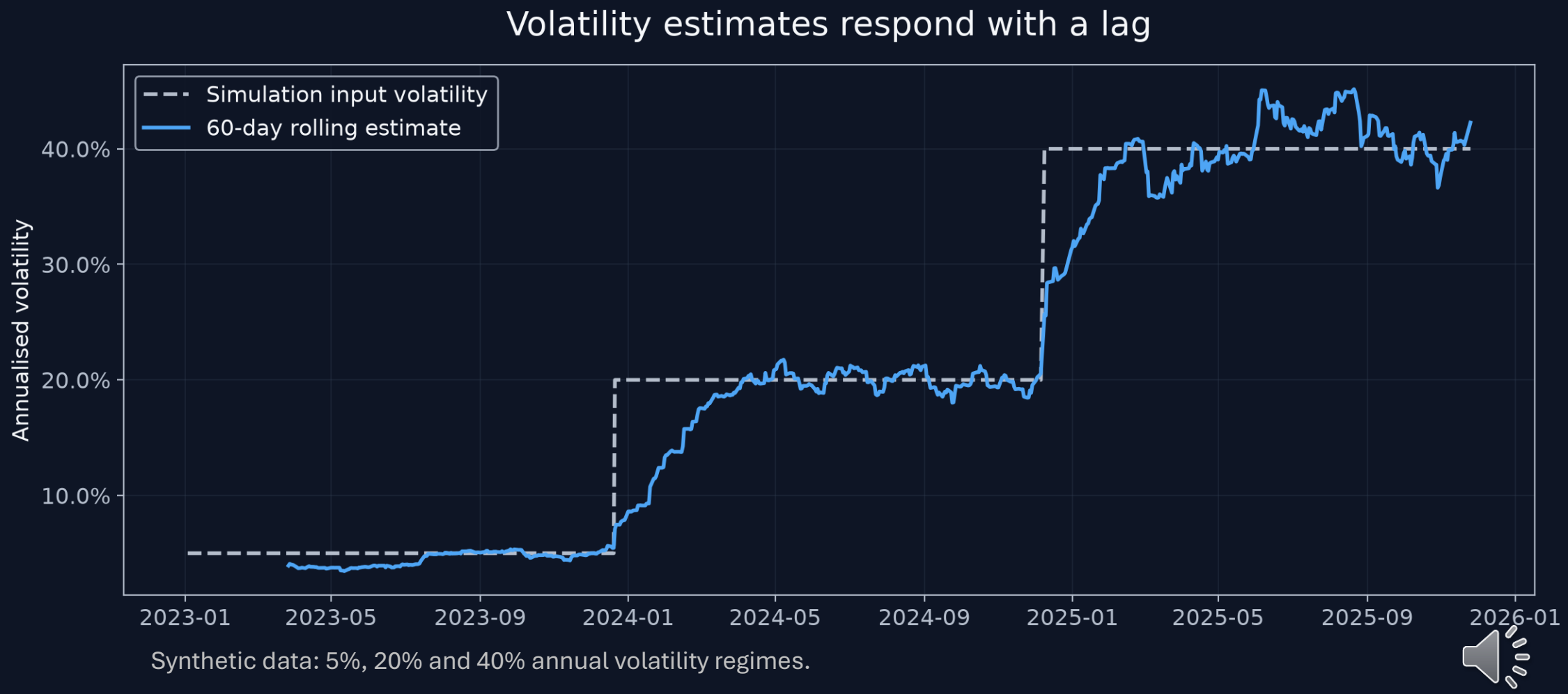
Annualised volatility ≈ Daily standard deviation × $\sqrt{252}$.

1.26% daily volatility × $\sqrt{252}$ ≈ 20% annual volatility

Assumes 252 trading days per year and square root of time scaling.



Volatility Estimates Respond with a Lag



Synthetic data: 5%, 20% and 40% annual volatility regimes.

Why Volatility Sizing needs Limits

Assume: Annual risk target = 2%, Signal = +1 (long)

Estimated annual volatility	Unconstrained target weight
20%	2% ÷ 20% = **10%**
10%	2% ÷ 10% = 20%
2%	2% ÷ 2% = 100%
1%	2% ÷ 1% = 200%

This is why we use exposure limits (for example: maximum absolute weight of 20% per asset).

A risk target does not replace exposure limits.



How Volatility Changes Position Size

The same signal produces different position sizes



The cap limits exposure when estimated volatility is low.

Gross Exposure, Net Exposure and Leverage

Assume portfolio equity of €100,000

Position	Signed Exposure	Weight
Long Stock A	+€80,000	+80%
Short Stock B	-€80,000	-80%

Gross exposure = €80,000 + €80,000 = €160,000
Net exposure = €80,000 - €80,000 = €0
Gross leverage = €160,000 ÷ €100,000 = 1.6×

Zero net exposure does not mean zero risk.



Correlation and Portfolio Risk

Stock A: 10% weight × 20% volatility = 2% standalone risk

Stock B: 5% weight × 40% volatility = 2% standalone risk

How much risk do they create together?

Correlation between returns	Estimated annual portfolio volatility
0 (Uncorrelated)	2.83%
+0.5 (Positively correlated)	3.46%
+1 (Perfectly positively correlated)	4%

Assumes fixed weights and the remaining 85% held in cash with zero volatility.

$$\sigma_p = \sqrt{(w_a\sigma_a)^2 + (w_b\sigma_b)^2 + 2\rho w_a w_b \sigma_a \sigma_b}$$

Position sizes must be assessed together, including correlations

Position and Concentration Limits

Check each proposed position against both individual and portfolio limits.

Limit	Illustrative rule	Purpose
Single position	Absolute weight ≤ 20%	Limit exposure to one instrument.
Sector concentration	Sector gross exposure ≤ 30% of equity	Limit related exposures
Portfolio gross leverage	Gross exposure ÷ Equity ≤ 1.5×	Limit total exposure

Example: 3 individual tech holdings of 15% fall within single position limits but breach the sector concentration limit.

Passing a position limit does not mean the portfolio passes its limits.



Liquidity, Turnover and Trading Costs

Reaching a target position involves execution costs and liquidity constraints.

Consideration	What to check
Liquidity	Order size relative to available depth and typical trading volume.
Turnover	How much buying and selling the strategy requires.
Trading costs	Commissions, bid-ask spread, slippage and market impact.

Example: Current holding: 190 shares | Target: 200 shares → Buy 10 shares
Trade value = 10 × €50 = €500.
(Assumed) Execution cost = €2.50: 0.5% of trade value.

Small target changes may not justify the trading costs.



Putting the Sizing Process Together

Return to our example (Stock A) and follow the process from signal to proposed trade.

Step	Running example
Signal	+1: Long
Size using volatility	2% risk target ÷ 20% volatility = 10% weight
Check portfolio constraints	Position, sector and leverage constraints pass.
Convert to shares	10% × €100,000 ÷ €50 = 200 shares
Calculate the required trade	200 target – 190 currently held = **Buy 10**
Check execution	€500 trade: Assumed cost €2.50.

The signal proposes exposure; risk and execution checks shape the final trade.



Understanding Drawdown

Drawdown measures how far portfolio equity has fallen below its previous peak.

Drawdown = (Previous peak – Current Equity) ÷ Previous peak

Stage	Portfolio Equity	Drawdown
Starting value	€100,000	0%
New peak	€110,000	0%
Subsequent decline	€99,000	10%
Partial recovery	€104,500	5%

Drawdown is measured from the peak, not starting capital.



Responding to Losses

Define loss-response rules before trading, including when and how trading resumes.

Control	Trigger	Possible response
Position stop-loss	A position reaches a predefined loss threshold.	Reduce or exit the position.
Portfolio drawdown rule	Equity falls a specific percentage below peak.	Reduce portfolio exposure or pause trading.
Operational kill switch	Faulty data, unexpected orders or a system failure	Block new orders and follow a recovery procedure.

Example: At 10% drawdown, halve portfolio risk targets and investigate.

A loss threshold triggers an action, it does not guarantee a maximum loss.



Stress-Testing the Portfolio

Stress-tests estimate what would happen under specified adverse scenarios.

Example: Portfolio Equity: €100,000 | Long A: €80,000 | Short B: €80,000

Hypothetical scenario	Estimated P&L
A drops 5%, B rises 5%	-€8,000
A falls 10%, B rises 10%	-€16,000
Both stocks fall 5%	€0 (before costs)

Static position: excludes execution costs, financing and changes in exposure.

Zero net exposure can conceal substantial adverse scenario losses.



Python: From Returns to Target Weights

```
returns = prices.pct_change()

vol = (
    returns.rolling(60, min_periods=60).std(ddof=1)
    * np.sqrt(252)
)

raw_weight = 1.0 * 0.02 / vol

capped_weight = raw_weight.clip(-0.20, 0.20)
```

Signal = +1 | Annual risk target = 2% | Target-weight cap = 20%



All methods remain in cash during the 60-day return warm up.

Python: From Targets to Trades

Monday after close: calculate target → Tuesday close: execute.

```
execution_targets = targets.shift(1).fillna(0.0)

# Inside the daily loop:
equity_before_cost = cash + shares * price
weight = execution_targets.loc[date]

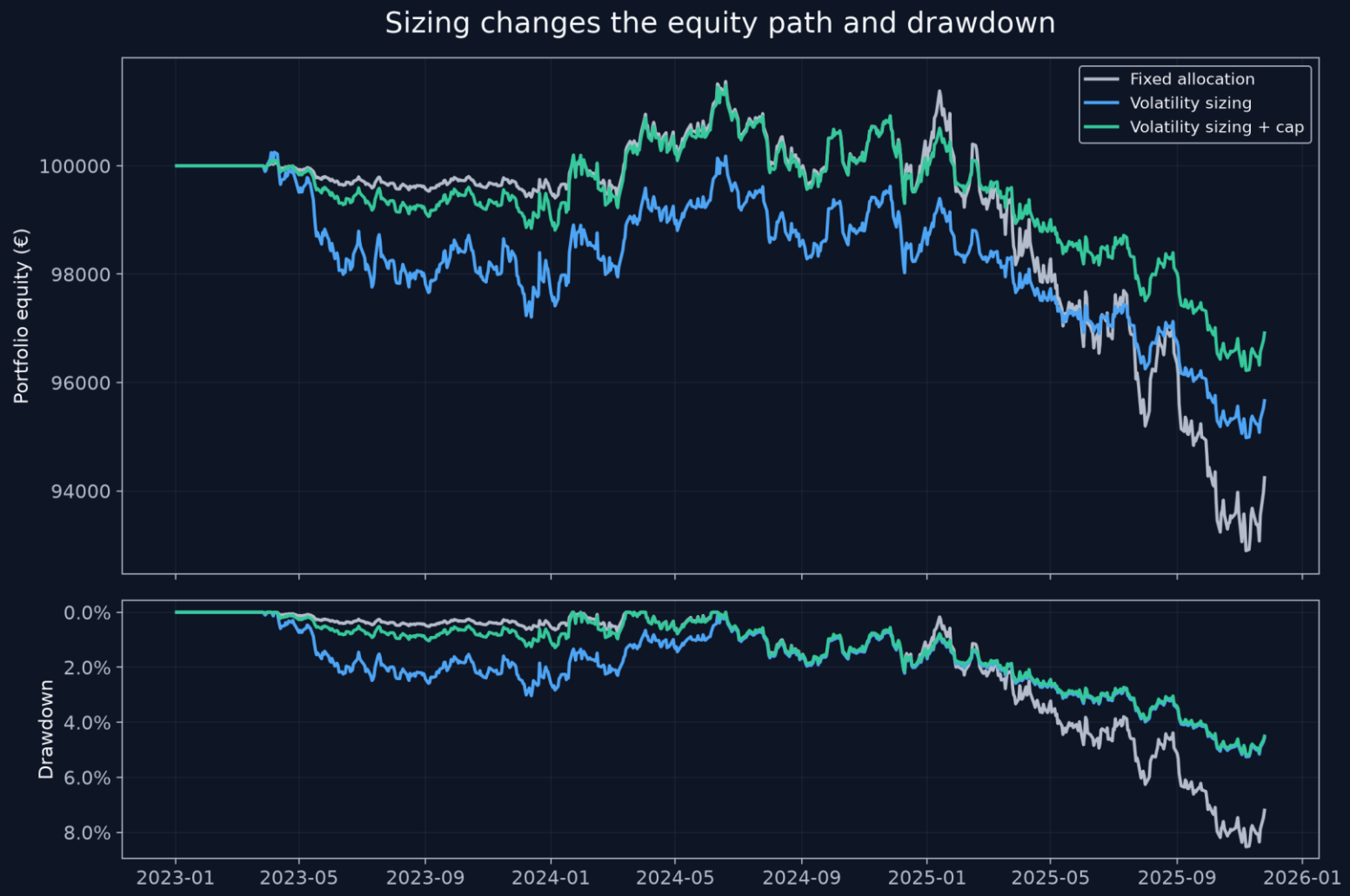
desired_shares = weight * equity_before_cost / price

trade_shares = desired_shares - shares
trade_value = trade_shares * price
cost = abs(trade_value) * COST_RATE
```

Trade the difference from actual holdings and charge costs on traded value.



Same Signal, Different Risk Paths



Synthetic data; constant long signal; returns after assumed execution costs.

Risk and Trading Costs: Results

Method	Net return	Annual volatility	Max drawdown	Annual turnover
Fixed 10%	-5.76%	2.78%	8.53%	30.53%
Volatility sizing	-4.33%	2.32%	5.26%	89.30%
Volatility + cap	-3.09%	2.01%	5.19%	37.34%

One synthetic path: exposures differ. Turnover counts purchases and sales, including initial entry.

Lower realised risk can come with greater turnover.



Common Implementation Mistakes

There are many common mistakes implementing sizing rules.

- Using information unavailable at the decision time.
- Mixing daily and annual volatility units.
- Confusing target holdings with trading quantities.
- Ignoring costs, partial fills or shorting constraints.
- Checking individual positions without aggregating portfolio risk.

A correct formula still needs correct timing, units and execution.



Key Takeaways

- Convert signals into explicit target positions.
- Scale exposure using estimated risk.
- Apply portfolio and execution constraints.
- Monitor losses and define responses in advance.

